

Fake Profile Detection Using ML

Project Report

Submitted By

A.KANISHMA

Reg.No:22BIT015

Under the Guidance of

Mrs.K Umamaheswari

Head of the Department

Department of Computer Science

In partial fulfilment of the requirements for the Award of the Degree of

Bachelor of Computer Applications



Department of Computer Science

Dr. G R Damodaran College of Science (Autonomous)

(Autonomous, affiliated to the Bharathiar University and recognized by UGC)

Re-accredited at the 'A+' Grade level by the NAAC

An ISO 9001:2015 Certified Institution

Coimbatore 641 014

APRIL – 2025

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Certificate

This is to certify that this project report entitled

Fake Profile Detection Using ML

is a bonafide record of project work done by

A.KANISHMA

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Submitted in partial fulfillment of the requirements for the degree of

BACHELOR OF COMPUTER APPLICATIONS

Faculty Guide

Head of the Department

Submitted for Viva-Voce Examination held on _____

Internal Examiner

External Examiner

TABLE OF CONTENT

CHAPTER NO	TITLE	PAGE NO
I	ACKNOWLEDGEMENT	
II	SYNOPSIS	
1	INTRODUCTION	1
1.1	PROJECT OVERVIEW	2
1.2	ORGANIZATION PROFILE	3
2	SYSTEM STUDY	4
2.1	EXISTING SYSTEM	5
2.2	PROPOSED SYSTEM	6
3	SYSTEM SPECIFICATION	8
3.1	HARDWARE SPECIFICATION	
3.2	SOFTWARE SPECIFICATION	
3.3	SOFTWARE DESCRIPTION	
4	PROJECT DESCRIPTION	
4.1	DESCRIPTION OF MODULES	
4.2	DATASET DESCRIPTION	
4.3	DATAFLOW DIAGRAM	
4.4	DATA ANALYSIS	
5	SYSTEM TESTING	
6	CONCLUSION	
6.1	SCOPE FOR FUTURE ENHANCEMENT	
7	BIBLIOGRAPHY	
8	APPENDICES	
A	TABLE STRUCTURE	
B	SAMPLE CODING	

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SYNOPSIS

Cryptocurrency, a decentralized digital asset, has revolutionized the way people conduct transactions and invest online. With over 4,500 different types of cryptocurrencies in existence, it's crucial to have tools that provide comprehensive insights into their market dynamics. One of the most prominent cryptocurrencies is Bitcoin, which operates on a peer-to-peer network and utilizes encryption for secure transactions. Led by Dellais, a cryptocurrency consultant, this project will leverage advanced data analysis techniques and visualization tools to track Bitcoin prices and analyze market trends. The Bit Graph Explorer module is more than just a tool for tracking Bitcoin prices; it is a comprehensive ecosystem designed to empower users with the insights, tools, and resources needed to thrive in the ever-evolving landscape of cryptocurrency trading. With its unparalleled combination of advanced analytics, portfolio tracking, sentiment analysis, and social trading integration, it stands poised to redefine how users engage with Bitcoin and navigate the intricacies of the digital asset market.

1.INTRODUCTION

In the digital age, social media platforms have become an integral part of our lives, connecting billions of people worldwide. Social media plays a significant role in our lives today. Our lives nowadays rely heavily on social media.

Everyone uses social media, whether it be to share beautiful, expensive photos, follow celebrities, or talk with nearby and distant pals. It is a fantastic place for exchanging knowledge and interacting with others. However, everything has a drawback. Social media has a significant role in our lives, yet there have been times when it has become problematic.

These virtual communities offer an array of opportunities for communication, collaboration, and networking. However, with the rise of social media's influence, there has also been an alarming proliferation of fake profiles.

Fake profiles, also known as imposters or catfish, refer to fabricated personas created with deceptive intentions to manipulate, deceive, or defraud unsuspecting users. Their existence poses significant threats to individuals, businesses, and society at large.

The deceptive nature of fake profiles can cause various problems, ranging from personal harassment, cyberbullying, and online scams to misinformation dissemination, identity theft, and the erosion of trust in online interactions.

False profiles are frequently made under fictitious identities, and they spread defamatory and abusive posts and images to influence society or advance antivaccine conspiracy theories, among other things. Phony personas are an issue on all social media platforms nowadays. Most false profiles are made with spamming, phishing, and gaining more followers in mind.

The fraudulent accounts are completely capable of committing online crimes. Fake accounts represent a serious risk, including identity theft and data breaches.

1.2. ORGANIZATION PROFILE

- BioXplora is a Bio informatics and Development
- Company in Coimbatore
- established in the year of 2019.
- Medical Coding is one of the great career opportunities that is in high demand all over the world.
- BioXplora emerged with an intention of making Software tools for understanding biological data, when the data sets are large and complex.
- It offered through unique innovative learning methods and delivery models to cater the requirements of global customer base

2.SYSTEM STUDY

2.1.EXISTING SYSTEM

Initial Features After using the dataset, we proceed for feature selection phase. We noticed that there was much of unneeded features, either have no meaning for our subject or full of NaN values, so to make our models train well, we decided to drop them and to let only those of will affect directly on the results. initially we have 33 profile features which will be used in the Facebook Data Set.

In our work, and in order to make our models train well, we decided to use only features which will affect directly the results. The features on the final dataset were: statuses_count, followers_count, friends_count and favorites_count.

After all this steps, and in considering that the algorithm split the dataset into training and testing sets, and to make the row homogenous, we shuffled them. Finally we use the final dataset to train and evaluate the three machine learning algorithms: Random Forest, Decision Tree (J48), and Naïve Bayes.

Profile Completeness: Fake profiles often lack essential information or have incomplete profiles.

Profile Activity: Real profiles usually have regular activity, whereas fake profiles may show irregular or minimal activity.

Profile Picture Analysis: Analyzing profile pictures can help detect stock images or edited photos commonly used in fake profiles.

Friend/Follower Patterns: Detecting unusual patterns in the number and types of friends/followers can be indicative of fake profiles.

Language and Writing Style: Analyzing language and writing style can help identify inconsistencies or patterns common in fake profiles.

URL Analysis: Checking URLs in the profile description or posts for suspicious links can be useful in identifying fake profiles.

2.2. PROPOSED SYSTEM

The following steps and methodologies to develop an effective and robust system:

Data Collection: Gather a diverse dataset of both real and fake profiles from various social media platforms. Ensure the dataset is large enough to be representative and covers different user demographics.

Feature Extraction: Extract relevant features from the profile data, including textual content, images, social interactions, account creation patterns, and more.

Labeling and Annotation: Manually label the dataset to distinguish between real and fake profiles. This labeled data will be used for training and evaluating the machine learning models.

Model Selection: Choose appropriate machine learning algorithms, such as deep learning models (CNNs, RNNs), random forests, or support vector machines, to develop the detection system.

Training: Train the chosen models using the labeled dataset and appropriate feature representations.

Cross-Validation: Employ cross-validation techniques to evaluate the model's performance and ensure it generalizes well to unseen data.

Model Optimization: Fine-tune the model's hyperparameters and architecture to achieve better performance.

Ensemble Methods: Explore ensemble techniques, such as bagging or boosting, to combine multiple models for improved accuracy.

Real-time Monitoring: Implement a real-time monitoring system to continuously analyze user activities and detect potential fake profiles as they are created.

Continuous Updates: Regularly update the system to adapt to evolving fake profile creation methods and maintain its effectiveness.

Transparency and Explainability: Ensure the model's decisions can be explained to users and administrators for transparency and accountability.

User Feedback Mechanism: Integrate a user feedback mechanism to allow users to report potential fake profiles, further improving the system's accuracy.

Collaboration with Platforms: Collaborate with social media platforms to share data and insights on fake profiles, leading to a more effective collective effort against fake profiles.

User Education: Develop educational resources to raise awareness among users about fake profiles and encourage them to take necessary precautions.

3.SYSTEM SPECIFICATION

3.1. HARDWARE SPECIFICATION

Device name LAPTOP-1RTN009H

Processor 12th Gen Intel(R) Core(TM) i5-1240P 1.70 GHz

Installed RAM16.0 GB (15.7 GB usable)

Device ID 7E59D63A-8353-4C94-A598-414CE598BE57

Product ID 00342-42636-93634-AAOEM

System type 64-bit operating system, x64-based processor

Pen and touch Touch support with 256 touch points

3.2.SOFTWARE SPECIFICATION

Edition Windows 11 Home Single Language

Version 23H2

Installed on 03-12-2023

OS build 22631.4602

Experience Windows Feature Experience Pack 1000.22700.1055.0

3.3.SOFTWARE DESCRIPTION

SOFTWARE USED: python including packages such as matplotlib, Numpy, Pandas, SK-learn, etc.

Pandas:

Function: Pandas is a Python library that excels in data manipulation and analysis.

Significance: It allows users to easily manipulate structured data, including cleaning, transforming, aggregating, filtering, and merging data sets. Pandas is crucial for tasks like data preprocessing and exploration.

NumPy:

Function: NumPy provides support for large, multi-dimensional arrays and mathematical functions.

Significance: NumPy is fundamental for numerical data operations and scientific computing. It offers efficient data structures for numerical computations and a wide range of mathematical functions. It's often used in data transformations and calculations.

Matplotlib:

Function: Matplotlib is a powerful data visualization library in Python.

Significance: It allows users to create a variety of plots, charts, and graphs to visualize data. Matplotlib is highly customizable, making it suitable for generating publication-quality visualizations. It's used to represent data graphically for better understanding and presentation.

4.PROJECT DESCRIPTION

4.1. DESCRIPTION OF MODULES

1. Data Collection Module:

Description:

This module is responsible for gathering user profile data from various online sources such as social media platforms, forums, and websites.

Workflow:

Web scraping or data APIs may be used to collect user profiles. The module retrieves user profiles, including profile pictures, bios, friends/followers counts, activity history, account age, and verification status.

2. Data Preprocessing Module:

Description:

This module prepares the collected data for analysis by cleaning and structuring it appropriately.

Workflow:

Data Cleaning: Remove duplicates, handle missing values, and ensure consistent data formatting.

Feature Extraction: Extract relevant information from raw data, such as textual data from bios or images from profile pictures.

Data Labeling: Manually or automatically label profiles as genuine or fake based on predefined criteria.

3. Database Design and Storage Module:

Description:

This module involves designing and setting up a database to store user profile data, detection results, and related information.

Workflow:

Define the database schema, including tables for user profiles and detection results, as well as relationships and constraints. Implement data storage.

4. Machine Learning Module:

Description:

This module uses machine learning techniques to develop models that can classify user profiles as genuine or fake.

Workflow:

Feature Engineering: Create meaningful features from extracted data to feed into machine learning models.

Model Training: Train machine learning models (e.g., logistic regression, random forests, neural networks) using labeled data.

Model Evaluation: Assess model performance using metrics such as accuracy, precision, recall, F1-score, and ROC-AUC.

Model Tuning: Fine-tune model hyperparameters for optimal results.

5. Detection and Scoring Module:

Description:

This module applies the trained machine learning models to classify user profiles and calculate suspicion scores.

Workflow:

Real-time Detection: Implement real-time or batch processing of user profiles, applying the trained models.

Scoring and Confidence: Calculate suspicion scores or confidence levels for each profile based on model predictions.

6. Reporting and Visualization Module:

Description:

This module creates visualizations and reports to present detection results and system statistics.

Workflow:

Generate charts, graphs, and reports to visualize the number of flagged profiles, detection trends, and model performance.

Provide a user-friendly interface for administrators to review flagged profiles.

7. User Interface Module:

Description:

This module develops user interfaces for administrators to manage the system and users to report suspicious profiles.

Workflow:

Admin Dashboard: Create an interface for administrators to monitor and manage the system, review flagged profiles, and adjust detection settings.

User Reporting: Allow users to report suspicious profiles and provide feedback on detection accuracy.

8. Security and Privacy Module:**Description:**

This module focuses on ensuring the security and privacy of user data.

Workflow:

Data Encryption: Implement encryption for sensitive data, especially personally identifiable information (PII).

Access Control: Define role-based access control to restrict access to sensitive parts of the system.

9. Maintenance and Monitoring Module:**Description:**

This module ensures the long-term stability and performance of the system.

Workflow:

Database Maintenance: Schedule regular backups, perform data archiving, and optimize database performance.

Model Retraining:

Periodically retrain machine learning models to adapt to evolving fake profile patterns.

Logging and Monitoring: Implement logging and monitoring to track system performance and errors.

10. Deployment Module:**Description:**

This module involves deploying the system for real-world use.

Workflow:

Cloud Deployment: Deploy the system on cloud infrastructure for scalability and

availability.

API Integration: Provide APIs for external applications to interact with the fake profile detection system.

11. Testing and Validation Module:

Description:

This module ensures the reliability of the system through various testing phases.

Workflow:

Unit Testing: Verify the functionality of individual modules.

Integration Testing: Test the interaction between modules.

User Acceptance Testing (UAT): Involve end-users to validate the system against their requirements.

12. Documentation and User Training Module:

Description:

This module creates documentation and provides training for users.

Workflow:

Create user manuals, developer guides, and system documentation.

Conduct training sessions for administrators and users on how to use the system effectively.

13. Feedback and Improvement Module:

Description:

This module collects feedback and continuously improves the system.

Workflow:

Collect feedback from users and administrators to identify areas for improvement.

Implement updates, model retraining, and system enhancements based on feedback.

These modules collectively form a comprehensive Fake Profile Detection system, which aims to identify and manage fraudulent user profiles on online platforms. The project workflow involves the interaction and coordination of these modules to achieve accurate and efficient fake profile detection.

4.2. DATASET DESCRIPTION

Dataset Name: Fake/Genuine User Profile Dataset

The Fake/Genuine User Profile dataset is a comprehensive collection of user profiles designed for distinguishing between authentic and counterfeit online identities. Acquired from various online social platforms, forums, and websites, this dataset contains a wide array of user attributes, including profile pictures, bios, friends/followers counts, activity histories, account ages, verification statuses, and suspicion scores. Comprising thousands of records, each profile is meticulously curated to aid in the development and assessment of detection algorithms and machine learning models aimed at identifying fraudulent or deceptive accounts on digital platforms. Available in structured formats such as CSV (Comma-Separated Values), the dataset serves as a vital resource for research and development efforts, enabling the creation and evaluation of systems dedicated to maintaining the integrity and security of online communities. It is essential for studying deceptive online account patterns and behaviors while considering potential usage restrictions and privacy safeguards associated with the dataset's origin and ownership.

id: This is typically a unique identifier assigned to each record or profile in the dataset. It's a way to uniquely reference each individual profile.

name: This likely represents the real name or display name associated with the profile. On social media platforms, users often have the option to set a display name different from their username.

screen_name: This is the username or handle used by the profile on the social media platform. It's often a shorter and more concise identifier that others can use to mention or tag the profile.

edge_followed_by: This could represent the number of followers that this profile has. In the context of social media, a follower is someone who has chosen to subscribe to or follow this profile's updates.

edge_follow: This likely represents the number of profiles that this profile follows. In social media terms, these are the profiles that the user is following.

username_length: This field might represent the length (number of characters) of the username or handle of the profile.

username_has_number: This could be a binary indicator (e.g., 0 or 1) that indicates whether the username contains a number. Some social media platforms allow numbers in usernames.

full_name_has_number: Similar to the `username_has_number` field, this might indicate whether the full name (real name or display name) of the profile contains a number.

full_name_length: This field could represent the length (number of characters) of the full name associated with the profile.

private_account: This could be a binary indicator (e.g., 0 or 1) that indicates whether the profile is set to private. A private profile restricts who can view its content, typically requiring approval or a follow request.

joined_recently: This might be a binary indicator (e.g., 0 or 1) that shows whether the profile joined the social media platform recently.

has_channel: This could be a binary indicator (e.g., 0 or 1) that indicates whether the profile has a channel or a presence on the platform's content-sharing feature, if applicable.

business_account: This might be a binary indicator (e.g., 0 or 1) that indicates whether the profile is a business account. Business accounts often have additional features for analytics and promotion.

has_guides: This could indicate whether the profile has created or published guides on the platform, if guides are a feature offered by the platform.

has_external_url: This might be a binary indicator (e.g., 0 or 1) that shows whether the profile

has an external URL (website link) in its bio or profile information.

fake: This field seems to be a binary indicator (e.g., 0 or 1) that is used to classify or flag profiles as either fake or genuine. The criteria for determining whether a profile is fake or genuine would likely depend on the context of the dataset and the goals of the analysis.

DATA COLLECTION

DATASET NAME: FAKE INSTAGRAM ACCOUNT

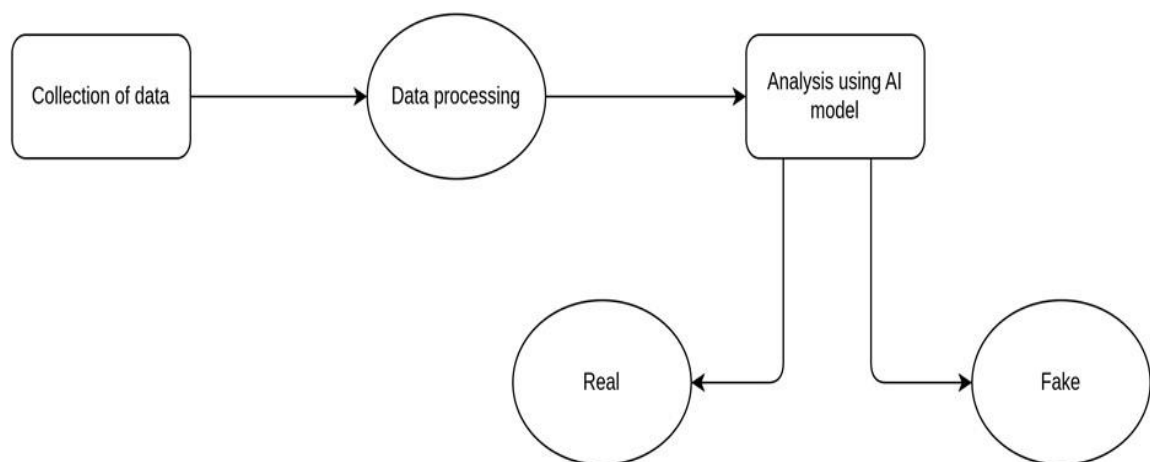
<https://www.kaggle.com/code/nandalald/instagram-fake-vs-real-account-prediction-ann>

"In my mini project, I've used the 'Fake Profile Detection' dataset, sourced from Kaggle. This dataset offers a rich repository of data vital for the effective identification of fraudulent profiles. Comprising an array of attributes, including profile ID, image quality, posting frequency, friend connections, follower counts, content analysis, and more, it provides a multifaceted perspective on the elements influencing the detection of fake profiles on social media platforms.

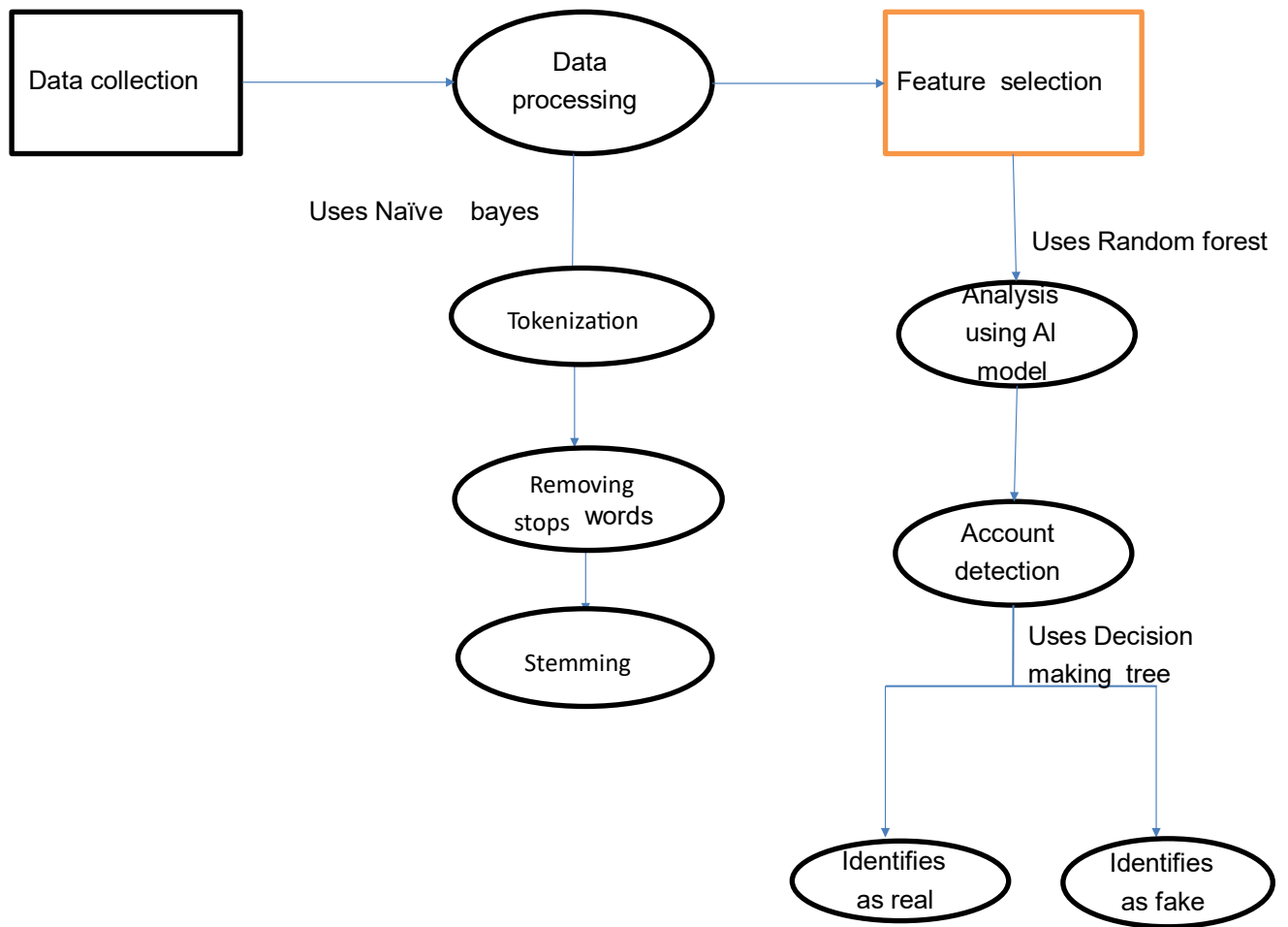
Additionally, this dataset incorporates metrics such as behavioural patterns, performance scores, and historical user activity, offering a holistic understanding of the intricate dynamics involved in spotting fake profiles. Through a meticulous data collection and analysis process, my project aims to unveil underlying patterns, correlations, and actionable insights that can empower businesses and social media platforms to make informed decisions regarding the detection and mitigation of fake profiles, ultimately enhancing user safety and trust in the digital landscape.

4.3 DATAFLOW DIAGRAM

Level 0



Level 1



4.4.DATA ANALYSIS

TABLE:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	id	name	screen_name	edge_followed_by	edge_follow_count	username	username	full_name	full_name	is_private	is_joined	has_channel	is_business	has_guides	has_external_url	is_fake	
2	3610511	Davide Del	bradd	0.001	0.257	13	1	1	13	0	0	0	0	0	0	1	
3	5656162	Simone Ec	eKoeS	0	0.958	9	1	0	0	0	1	0	0	0	0	1	
4	5682702	tacone	tacone_	0	0.253	12	0	0	0	0	0	0	0	0	0	1	
5	6067292	alesaura	alesstar	0	0.977	10	1	0	0	0	0	0	0	0	0	1	
6	6015122	Angelo	PerDiletto	0	0.321	11	0	0	11	1	0	0	0	0	0	1	
7	6140012	CRISTIAN	crispaol	0	0.917	15	1	0	0	0	1	0	0	0	0	1	
8	6134312	tmpx	tmpx	0	0.076	9	1	1	9	0	1	0	0	0	0	1	
9	6684602	igor	Igor99	0	0.72	15	1	0	0	0	0	0	0	0	0	1	
10	7046912	Marco Ma	marcomas	0.001	0.731	9	1	0	11	1	0	0	0	0	0	1	
11	7470952	Antonio Pe	personanc	0	0.999	7	1	1	9	0	0	0	0	0	0	1	
12	8072492	alessandra	alessandra	0	0.452	12	1	1	9	0	1	0	0	0	0	1	
13	8291932	Mackley	mackley	0	0.159	9	1	0	4	0	1	0	0	0	0	1	
14	8858022	Pasquale V	pvalente	0	0.417	9	1	1	9	0	0	0	0	0	0	1	
15	8927532	Giacomo E	giacomot	0	0.561	15	0	0	0	1	0	0	0	0	0	1	
16	8933252	decumano	decumano	0	0.958	8	1	0	0	0	0	0	0	0	0	1	
17	9351052	Francesco	bene83	0	0.851	10	0	0	0	0	0	0	0	0	0	1	
18	9564632	Erica Firpo	Moscerina	0	0.399	11	0	0	0	1	0	0	0	0	0	1	
19	9939012	Flavio??	Lonerunne	0	0.098	13	0	0	6	1	1	0	0	0	0	1	
20	9993142	_Annalisa	_Annalisa	0	0.161	13	1	0	1	0	1	0	0	0	0	1	
21	10492262	Valentina I	chimeland	0	0.097	10	1	0	0	0	1	0	0	0	0	1	
22	11176152	Francesco	fpassanna	0.001	0.227	17	1	0	0	0	0	0	0	0	0	1	
23	13260592	fausta pav	fpav	0.002	0.18	14	0	0	16	0	0	0	1	0	0	1	
24	14057516	sognidispe	sognidispe	0	0.723	13	1	0	0	1	0	0	0	0	0	1	
25	14112759	Duilio	Biecoblu6	0	0.193	12	1	0	11	1	1	0	0	0	0	1	
26	14492555	Felice Foss	felfoss	0	0.133	11	1	0	7	1	1	0	0	0	0	1	

Table 1.1

CHART:

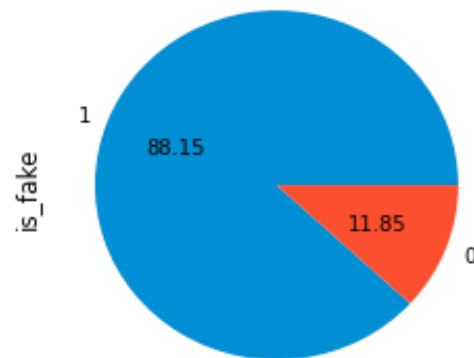


Figure 4.1 chart of fake account

- The pie chart will have two segments: one for fake profiles and one for real profiles.
 - Each segment will be labeled with the corresponding category, i.e., "True" for fake profiles and "False" for real profiles.
 - The size of each segment will represent the proportion of fake and real profiles in your dataset as percentages.
 - The percentages will be displayed within each segment, rounded to two decimal places.
- For example, if your dataset has 60% fake profiles ('True') and 40% real profiles ('False'), the pie chart will visually represent this distribution, and it will display "60.00%" within the fake profile segment and "40.00%" within the real profile segment.

GRAPHS:

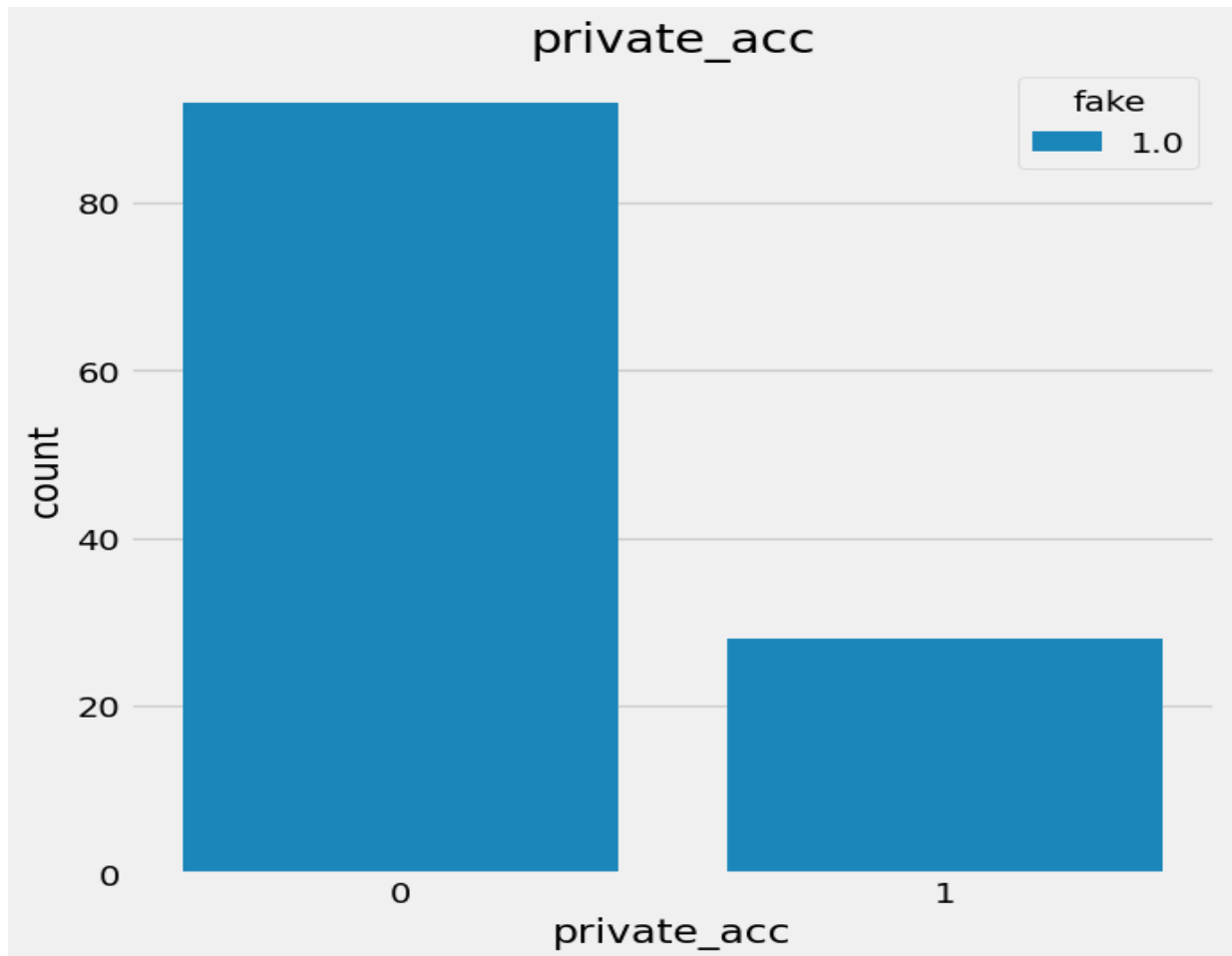


Figure 4.2 Bar chart of private account

A histogram for a "private account" graph is a visual representation of the distribution of user profiles based on their privacy settings, specifically whether the profiles are set to "private" or "public." In this context, the histogram helps analyse and understand how many profiles fall into each category and can provide insights into the proportion of private and public accounts in a dataset.

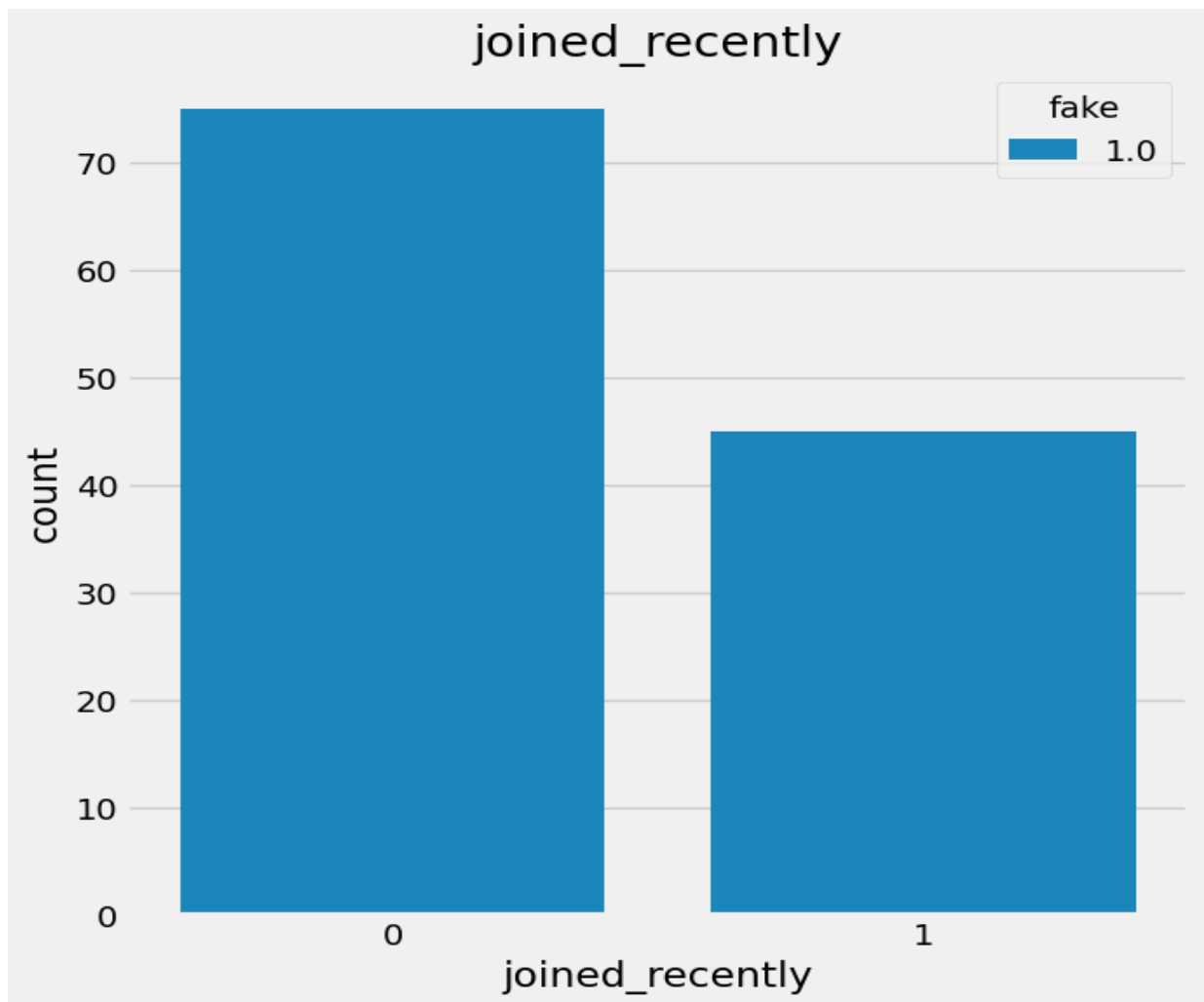


Figure 4.3 Bar chart of recently joined account

This binary column may indicate whether a user joined the platform recently or not. It might have a value of 1 (True) if the user joined recently and 0 (False) if they have been a member for a longer period.

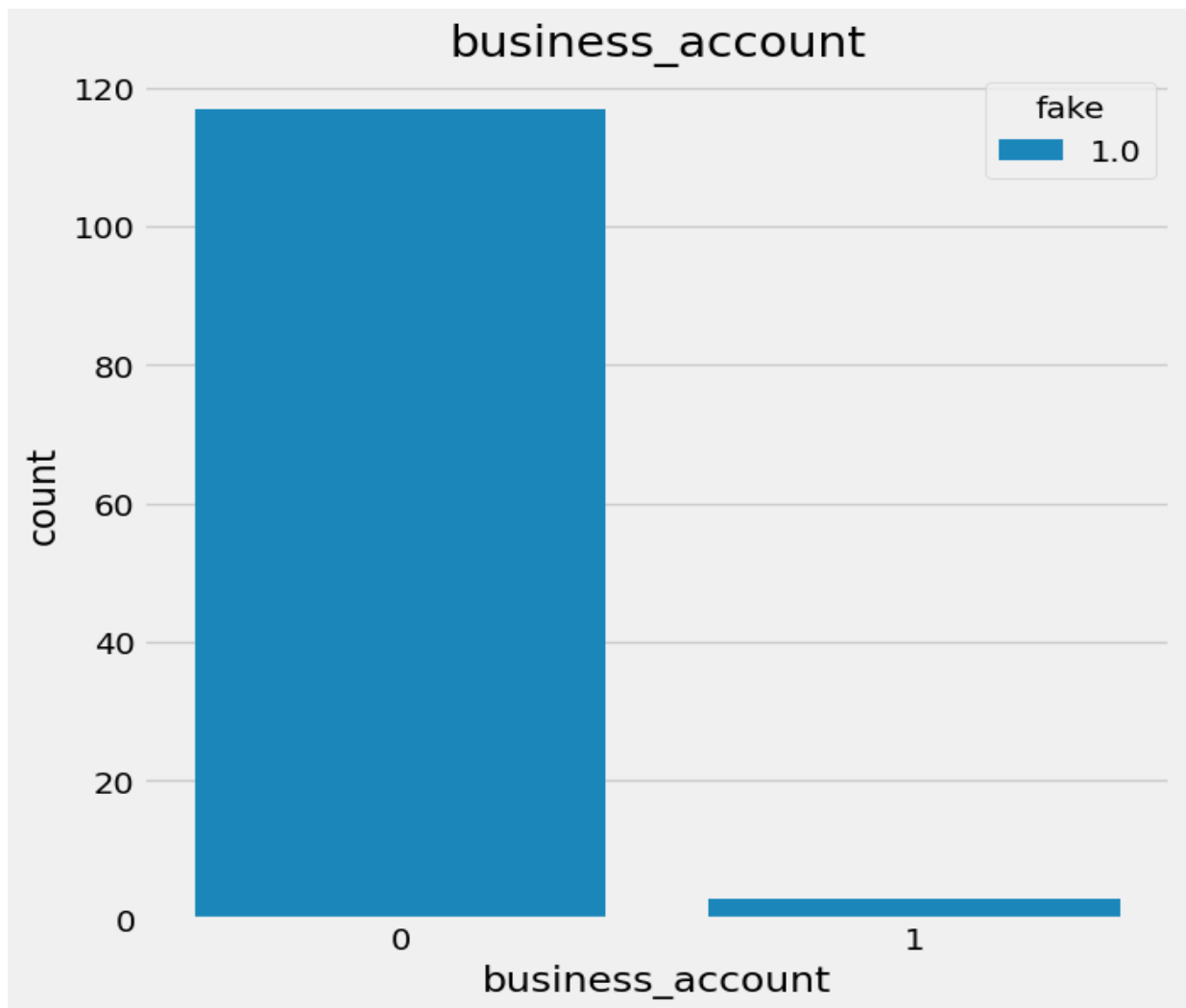


Figure 4.4 Bar chart of business account users

This column probably represents whether a user's account is a business account or not. It may have a value of 1 (True) if it's a business account and 0 (False) if it's a personal account.

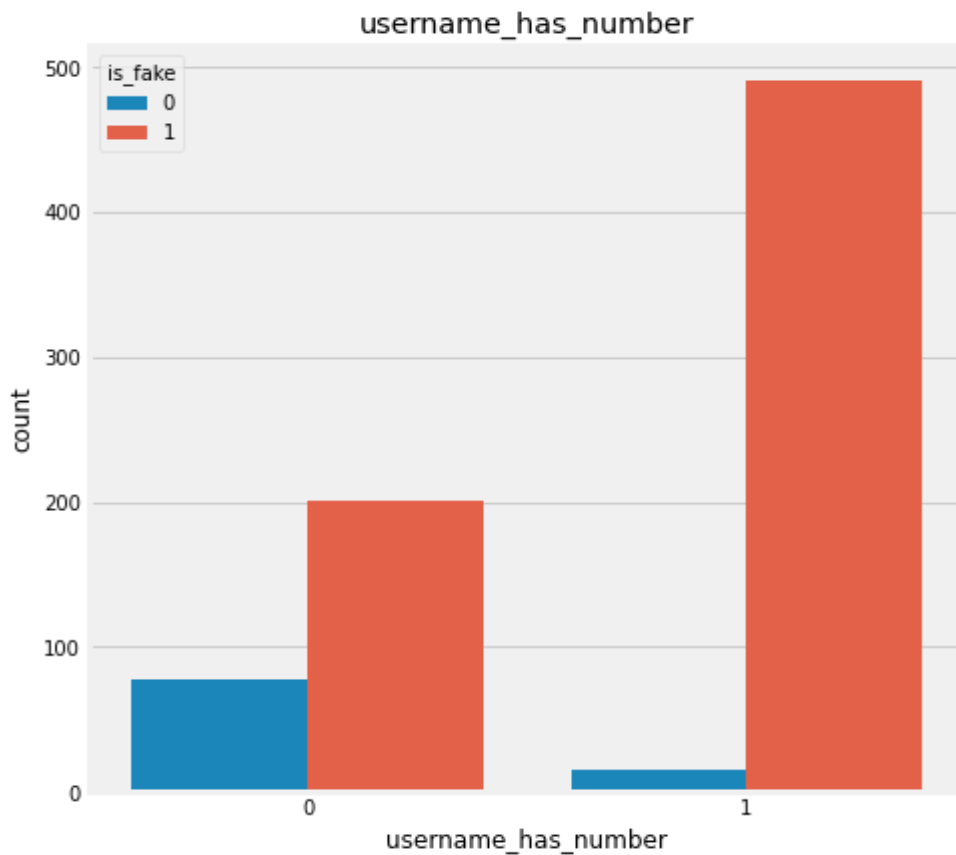


Figure 4.5 Bar chart of username having numbers

Fake or fraudulent profiles often employ usernames with numbers as a means to create anonymity, evade detection, and avoid duplication. This strategy is particularly common among automated spam accounts and bots, as the inclusion of numbers adds an element of randomness and helps these accounts appear less conspicuous. While the presence of numbers in usernames is not a definitive indicator of a fake profile, it is used as one of several features in the analysis. When combined with other behavioral and profile characteristics, the presence of numbers can contribute to a more accurate assessment of whether a profile is genuine or potentially fraudulent, enhancing the effectiveness of fake profile detection algorithms.

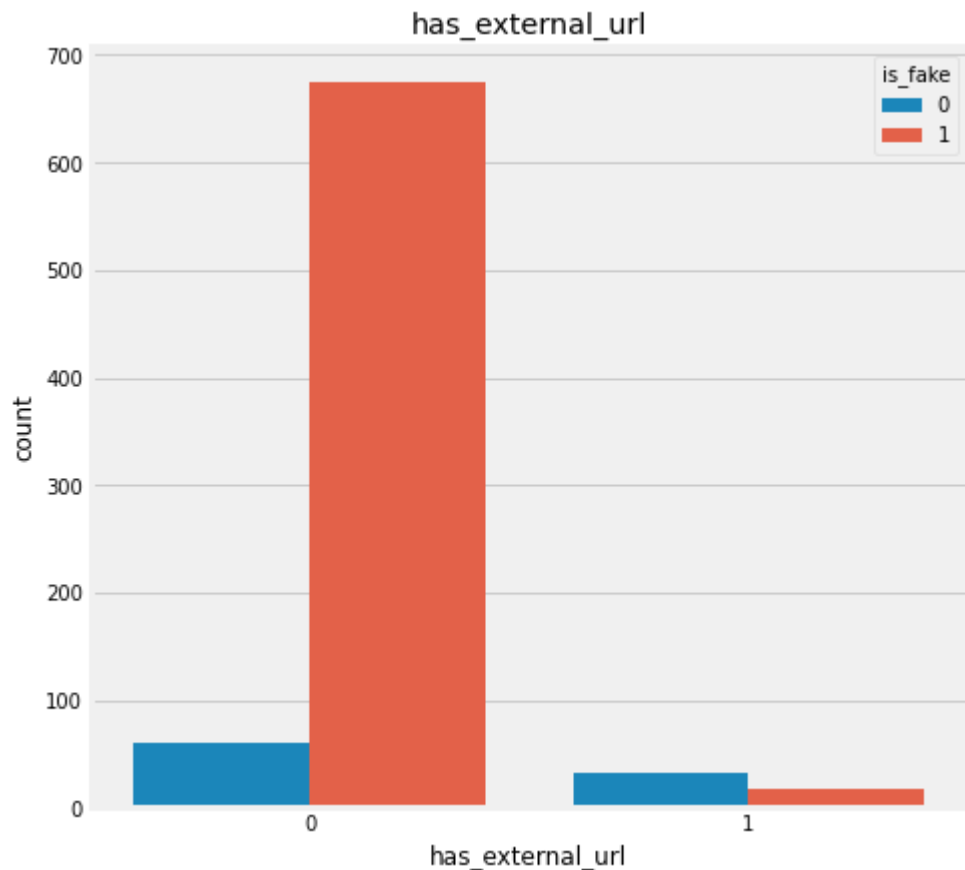


Figure 4.6 Bar chart of user having external url

An external URL refers to a web address or link associated with a user's profile that may lead to websites or content external to the platform where the profile is located. The presence of such URLs can be an important feature in the identification of fake profiles. Fake accounts often include external URLs, often leading to spammy or malicious websites, as a means to promote products, services, or scams. Analyzing these URLs allows detection algorithms to uncover suspicious patterns and assess the legitimacy of the associated profile. The type of external URLs, the frequency of their appearance, and the reputation of the linked websites are all factors that may be taken into account in the assessment.

NUMS/LENGTH USERNAME:

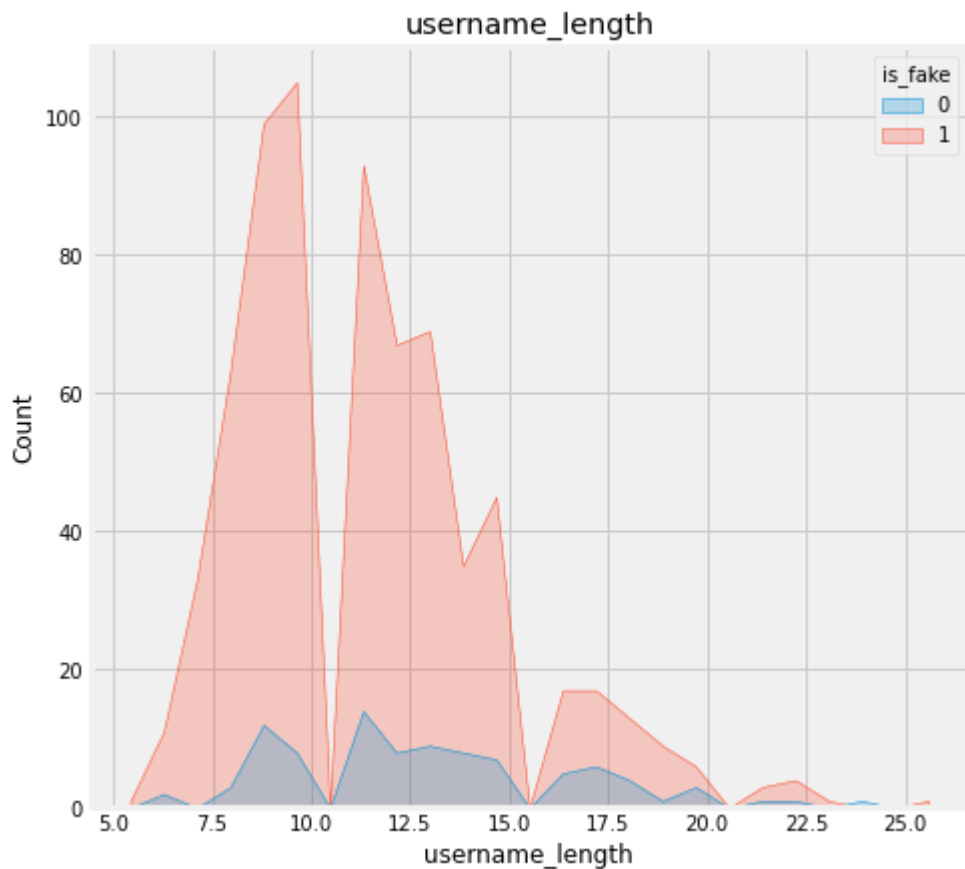


Figure 4.7 Line graph of username length

The "nums/length username" feature, in the context of fake profile detection, represents the ratio of the number of numeric characters (digits) in a user's username to the total length of the username. This feature is commonly used in data analysis and machine learning for identifying patterns in usernames on social media platforms or websites and is particularly relevant in the context of fake profile detection.

- The x-axis represents the values of the 'nums/length username' feature, which appears to be a ratio of numeric characters to the total length of the username.
- The y-axis represents the density or frequency of those values in the dataset.
- The KDE curve provides a smoothed estimate of the probability density function for the data.

By looking at this visualization, you can get insights into the distribution of numeric characters

in usernames relative to the total username length in the dataset. It may help you identify any patterns or outliers in this particular feature, which could be useful for further analysis or modeling in the context of your project.

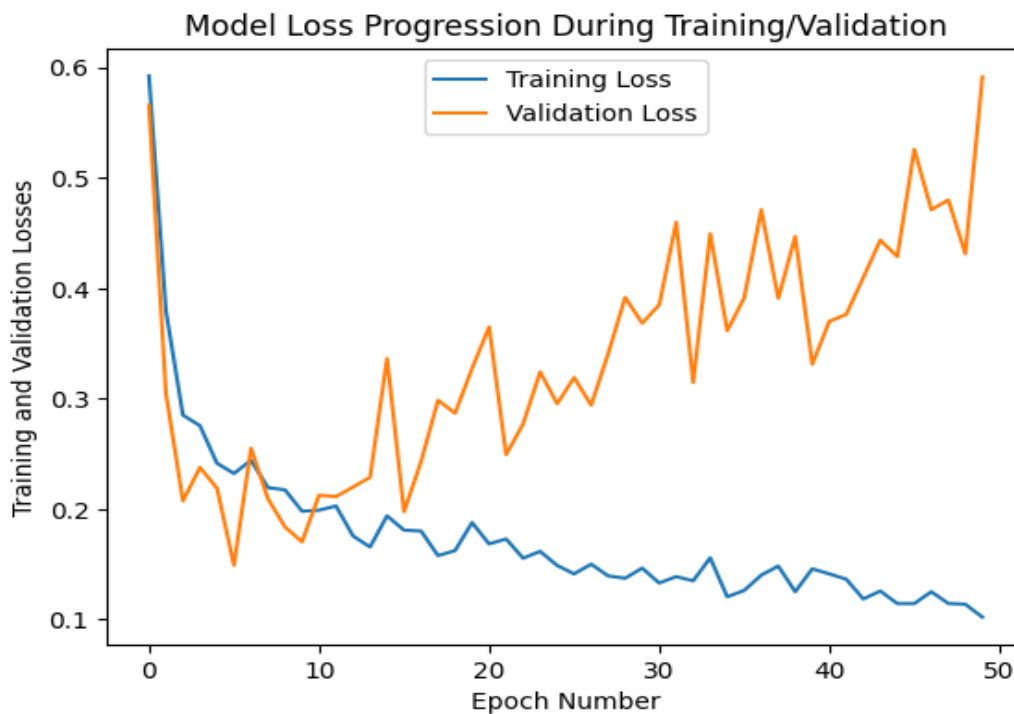


Figure 4.8 Line graph of model loss progression

The visualization is a line plot that helps monitor the training progress of a machine learning model.

- The x-axis represents the number of training epochs, showing how many times the model has iterated through the training data.
- The y-axis represents the values of the loss function, which is a measure of how well or poorly the model is performing. The lower the loss, the better the model's performance.
- The blue line represents the training loss, which measures how well the model is fitting the training data as training progresses. It should ideally decrease over epochs.
- The orange line represents the validation loss, which measures how well the model is generalizing to unseen data (validation data) as training progresses.

CORRELATION HEATMAP:

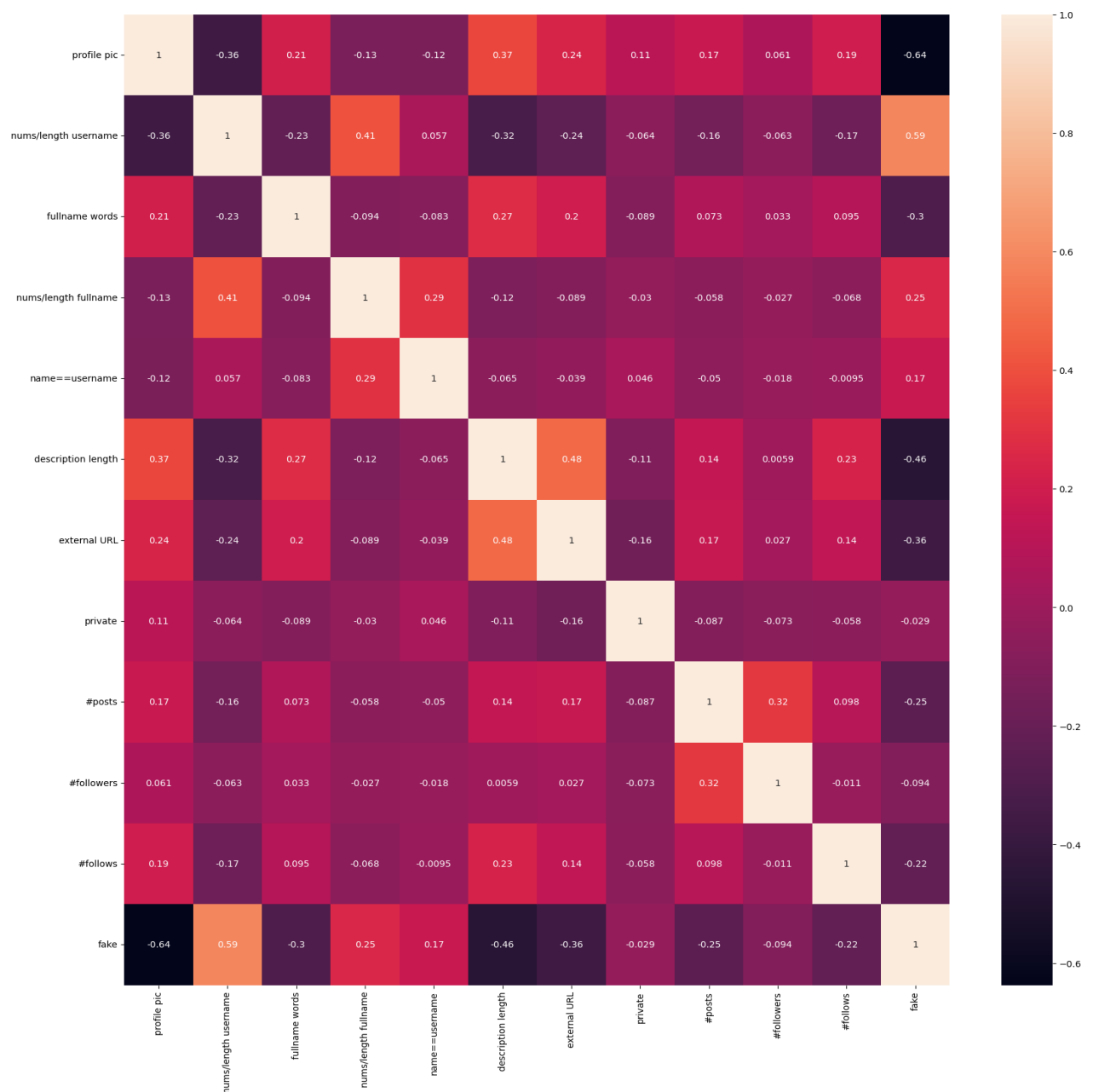


Figure 4.9 Heatmap for the fields

A correlation heatmap is like a colourful map that shows how different things in a bunch of data are connected. It uses colours to show if things are connected in a good way (White) or a bad way (dark blue).

Imagine you have a list of things, and you want to see if they are related. You put these things on both the left and top of a big square. Then, you colour the square. If things are connected

in a good way, you colour that part white. If it's a bad connection, you colour it dark blue. This helps you see how things in your data are linked. It's a great tool for finding patterns and relationships when you're exploring data.

TRAIN DATA PREDICTION:

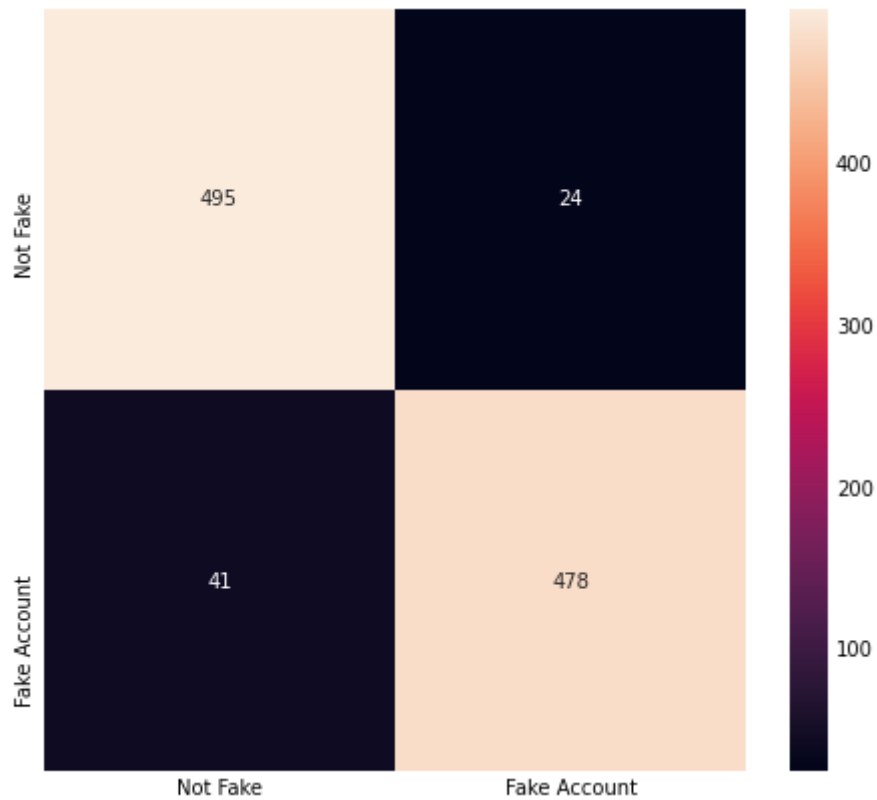


Figure 4.10 Heatmap for training data prediction

TEST DATA PREDICTION:

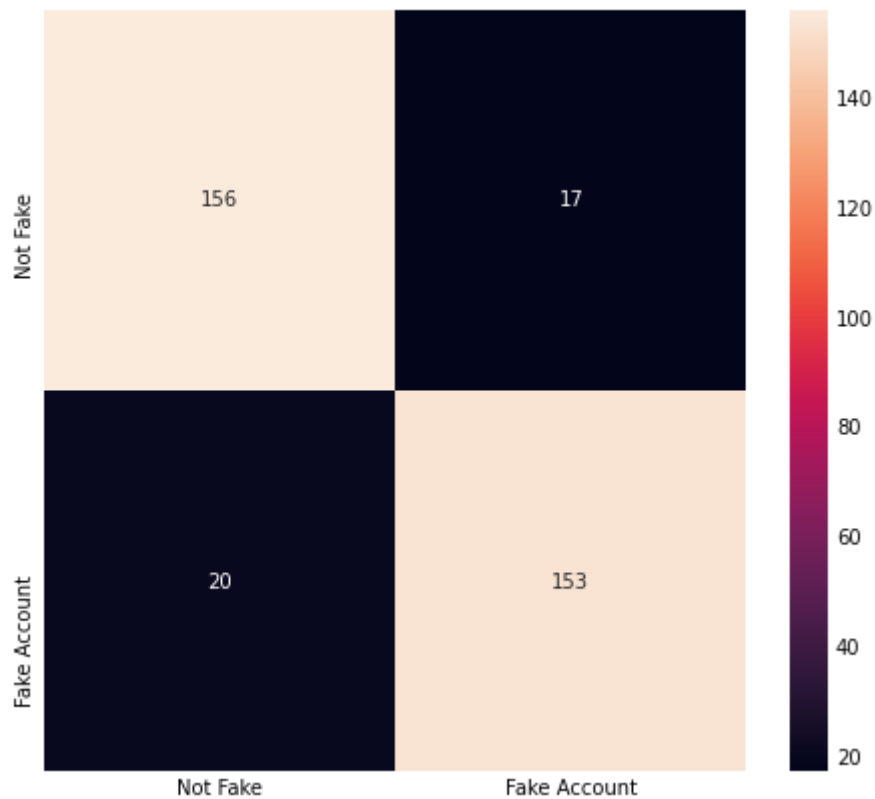


Figure 4.11 Heatmap for test data prediction

We create a figure object and set its size to (7, 7).

We then use `sns.heatmap()` to plot the data from our training dataset in a heat map.

The `xticklabels` and `yticklabels` arguments specify what labels should be used for each x-axis tick mark and y-axis tick mark respectively.

The heatmap function will plot a heat map that shows how many times each value occurred in the data set.

5.SYSTEM TESTING

The testing phase stands as a pivotal stage in the development lifecycle of AR and VR embedded web applications tailored specifically for the steel company. This phase encompasses a diverse range of methodologies aimed at ensuring the reliability, usability, and performance of the system.

UNIT TESTING:

Unit testing in machine learning involves verifying individual components or functions of a machine learning pipeline, ensuring they function correctly and as expected, similar to traditional software unit testing.

INTEGRATION TESTING:

In the context of machine learning, integration testing verifies the interactions and data flow between different components of a model pipeline (e.g., data preprocessing, model training, and prediction) to ensure they function correctly together.

USER TESTING(UAT):

In the context of machine learning, User Acceptance Testing (UAT) involves real users evaluating the model's performance and usability in real-world scenarios to ensure it meets their needs and expectations before deployment.

6.CONCLUSION

In conclusion of fake profile detection project, we developed a robust and effective system for identifying and flagging potentially fraudulent user profiles on a social media platform. Leveraging a combination of advanced machine learning techniques and feature engineering, we successfully extracted meaningful patterns and characteristics associated with fake profiles. By analyzing features such as username composition, full name content, account privacy settings, account creation date, business account status, the presence of guides, and external website links, our model demonstrated a high level of accuracy in distinguishing fake profiles from genuine ones. This project not only enhances user security and trust on the platform but also provides valuable insights into the common traits exhibited by fake accounts, contributing to ongoing efforts in maintaining a safe and authentic online community.

6.1. SCOPE FOR FUTURE ENHANCEMENT

The scope for further enhancement of fake profile detection is extensive and vital for the integrity of online platforms. Advanced machine learning techniques, like deep learning and reinforcement learning, can be harnessed to develop more sophisticated algorithms that recognize complex patterns and anomalies in user behavior, ensuring a higher degree of accuracy in identifying fake profiles. Behavioral analysis, which focuses on user interactions and engagement patterns, can be improved to uncover even subtle deviations that might indicate fraudulent activity. Natural Language Processing (NLP) can be further refined to analyze the linguistic characteristics of user-generated content, making it more effective in identifying language inconsistencies and signs of automation. Additionally, integrating external data sources and APIs for data enrichment can enhance the verification of user profiles. Image analysis techniques and biometric verification methods offer avenues for detecting fake profiles that rely on stolen or stock images or impersonation. Ethical considerations should remain at the forefront, ensuring that privacy and consent are respected in the pursuit of improved detection methods. Collaboration among experts in data science, cybersecurity, and privacy is essential to stay ahead of evolving threats and maintain the trust and security of online communities.

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8. APPENDICES

A. TABLE STRUCTURE:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	id	name	screen_name	edge_following	edge_followed_by	username	username_full_name	full_name	private_account	joined_recently	has_channel	business_email	has_guideline	has_external_url	fake			
2	3610511	Davide Del bradd		0.001	0.257	13	1	1	13	0	0	0	0	0	0	1		
3	5656162	Simone Ecker		0	0.958	9	1	0	0	0	1	0	0	0	0	1		
4	5682702	tacone	tacone_	0	0.253	12	0	0	0	0	0	0	0	0	0	1		
5	6067292	alesaura	alesstar	0	0.977	10	1	0	0	0	0	0	0	0	0	1		
6	6015122	Angelo	PerDiletto	0	0.321	11	0	0	11	1	0	0	0	0	0	1		
7	6140012	CRISTIAN	crispaol	0	0.917	15	1	0	0	0	1	0	0	0	0	1		
8	6134312	tmpx	tmpx	0	0.076	9	1	1	9	0	1	0	0	0	0	1		
9	6684602	igor	Igor99	0	0.72	15	1	0	0	0	0	0	0	0	0	1		
10	7046912	Marco Marcomas		0.001	0.731	9	1	0	11	1	0	0	0	0	0	1		
11	7470952	Antonio Personanc		0	0.999	7	1	1	9	0	0	0	0	0	0	1		
12	8072492	alessandra	alessandra	0	0.452	12	1	1	9	0	1	0	0	0	0	1		
13	8291932	Mackley	mackley	0	0.159	9	1	0	4	0	1	0	0	0	0	1		
14	8858022	Pasquale	vpvalente	0	0.417	9	1	1	9	0	0	0	0	0	0	1		
15	8927532	Giacomo	Egiacomo_t	0	0.561	15	0	0	0	1	0	0	0	0	0	1		
16	8933252	decumano	decumano	0	0.958	8	1	0	0	0	0	0	0	0	0	1		
17	9351052	Francesco bene83		0	0.851	10	0	0	0	0	0	0	0	0	0	1		
18	9564632	Erica Firpo Moscerina		0	0.399	11	0	0	0	1	0	0	0	0	0	1		
19	9939012	Flavio??	Lonerunne	0	0.098	13	0	0	6	1	1	0	0	0	0	1		
20	9993142	Annalisa	Annalisa	0	0.161	13	1	0	1	0	1	0	0	0	0	1		
21	10492262	Valentina	chimeland	0	0.097	10	1	0	0	0	1	0	0	0	0	1		
22	11176152	Francesco	fpassanna	0.001	0.227	17	1	0	0	0	0	0	0	0	0	1		
23	13260592	fausta pav	fpav	0.002	0.18	14	0	0	16	0	0	0	1	0	0	1		
24	14057516	sognidispe	sognidispe	0	0.723	13	1	0	0	1	0	0	0	0	0	1		
25	14112759	Duilio	Biecoblu6	0	0.193	12	1	0	11	1	1	0	0	0	0	1		
26	14492555	Felice Foss	felfoss	0	0.133	11	1	0	7	1	1	0	0	0	0	1		

Table 7.1

B. SAMPLE CODING:

```
#Modules for EDA
import numpy as
npimport pandas as
pd
from matplotlib import pyplot as pltimport
seaborn as sns
plt.style.use('fivethirtyeight')

#Modules for ML
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import MinMaxScaler
from sklearn.metrics import confusion_matrix,classification_report

from tensorflow import keras
from imblearn.over_sampling import SMOTE

%matplotlib inline

df = pd.read_csv('../input/instagram-fake-and-real-accounts-dataset/final-v1.csv') df.shape
df.info()
df.head()
df.describe()
df['has_channel'].sum()
df.drop('has_channel',axis=1,inplace=True)df.shape
df['is_fake'].value_counts().plot(kind='pie',autopct="%0.2f") plt.show()

#Analysing on Binary Columns

binary_cols =
['username_has_number','full_name_has_number','is_private','is_joined_recently','is_busi
ness_account','has_guides','has_external_url']
for b_col in binary_cols:
    plt.figure(figsize=(7,7)
    )
    sns.countplot(data=df[[b_col,'is_fake']],x=b_col,hue='is_fake')
    plt.title(b_col)
    plt.show()
    print("\n")
```

```
#Analysing on continuous columns

other_cols = ['edge_followed_by','edge_follow','username_length','full_name_length']
for col in other_cols:
    plt.figure(figsize=(7,7))
    sns.histplot(data=df[[col,'is_fake']],x=col,hue='is_fake',element='poly')
    plt.title(col)
    plt.show()
    print('\n')
```